

MELATONIN: THE BIOLOGY OF SLEEP

Introduction to Melatonin

Melatonin, often called 'hormone of darkness' is a natural hormone primarily produced by the pineal gland in the brain. It plays a crucial role in regulating the body's **sleep-wake cycle**, also known as the **circadian rhythm**. Often called the "sleep hormone," melatonin is secreted in response to darkness, helping signal the body that it is time to sleep. Production typically increases after sunset, peaks during the night, and falls in the early morning.

Melatonin is not only found in humans but is also present in animals, plants, and microbes, suggesting its vital and ancient role in life processes. Because of its wide range of actions, melatonin has been studied in relation to various conditions, including **insomnia**, **jet lag**, **seasonal affective disorder (SAD)**, and even **neurodegenerative diseases** like **dementia**.

In recent years, synthetic melatonin supplements have become popular as sleep aids, though their long-term effects are still being researched.

Melatonin and the Human Body

Melatonin is a hormone that deeply influences many functions in the human body, most notably the **regulation of sleep**. It is produced mainly by the **pineal gland**, a small, pea-shaped structure located near the center of the brain. The secretion of melatonin is controlled by the **light-dark cycle**; it increases in darkness and decreases in light.

Key Roles of Melatonin in the Human Body:

1. **Regulation of Sleep-Wake Cycle**
 - Melatonin helps synchronize the body's internal clock with the external environment (day and night).
 - It promotes sleep by making the body feel drowsy and preparing it for rest.
2. **Circadian Rhythm Control**
 - Circadian rhythms are 24-hour cycles that influence sleep, body temperature, hormone release, and digestion.
 - Melatonin helps maintain and adjust these rhythms, especially when they are disturbed (like during jet lag).
3. **Antioxidant Properties**
 - Melatonin acts as a powerful antioxidant, neutralizing harmful free radicals.
 - It helps protect cells and tissues from oxidative stress and damage.
4. **Immune System Support**
 - Melatonin strengthens immune responses and may improve the body's ability to fight infections and diseases.
5. **Mood Regulation**
 - Proper melatonin levels may help regulate mood and prevent mood disorders such as depression, especially seasonal depression (Seasonal Affective Disorder, or SAD).
6. **Potential Role in Aging and Neurodegenerative Diseases**
 - Studies suggest that melatonin might slow aging processes and could have protective effects against diseases like Alzheimer's and Parkinson's.

Melatonin and the Brain

Melatonin plays a major role in regulating brain functions, especially those related to **sleep, mood, and cognitive health**.

1. Production in the Brain

- Melatonin is mainly produced by the **pineal gland**, a tiny, pinecone-shaped gland located deep in the center of the brain, between the two hemispheres.
- Its production is controlled by the **suprachiasmatic nucleus (SCN)** of the hypothalamus, which acts as the brain's "master clock."
- The SCN receives information about light through the eyes and signals the pineal gland to start or stop melatonin production.

2. Regulation of Sleep

- Melatonin signals the brain that it is time to sleep.
- It binds to melatonin receptors in the brain to **lower brain activity** and **induce sleepiness**.
- It helps in **synchronizing** the brain's internal clock with the environment (light/dark cycle).

3. Effect on Mood and Mental Health

- Proper melatonin levels are linked to a **stable mood**.
- Low melatonin production is associated with mood disorders like **depression, anxiety, and seasonal affective disorder (SAD)**.

4. Cognitive Function and Brain Protection

- Melatonin acts as a **neuroprotective agent**.
 - It protects brain cells (neurons) from **oxidative stress** and **inflammation**, both of which can contribute to brain aging and diseases like **Alzheimer's** and **Parkinson's disease**.
 - Some research suggests that melatonin could help **slow down cognitive decline** in neurodegenerative conditions.
-

Melatonin and the Sleep-Wake Cycle

1. What is the Sleep-Wake Cycle?

The sleep-wake cycle, also known as the **circadian rhythm**, is the body's internal 24-hour clock that regulates periods of alertness and sleepiness. This cycle is controlled by a part of the brain called the **suprachiasmatic nucleus (SCN)**, located in the hypothalamus. The SCN responds to environmental cues, especially **light and darkness**, to maintain the rhythm.

2. Role of Melatonin in the Sleep-Wake Cycle

Melatonin is the **key hormone** that helps regulate the sleep-wake cycle. It is secreted by the **pineal gland** in response to darkness, and its levels are very low during the day.

- **At night:** Darkness triggers the SCN to send signals to the pineal gland to release melatonin. This signals to the body that it's time to wind down and prepare for sleep.
- **In the morning:** Light enters the eyes and inhibits melatonin production, making you feel alert and awake.

3. Secretion Pattern of Melatonin

Melatonin secretion follows a daily rhythm:

Time of Day	Melatonin Level
6:00 AM – 8:00 AM	Very low (near zero)
9:00 PM – 10:00 PM	Begins to rise
Midnight	High
2:00 AM – 3:00 AM	Peak levels
5:00 AM – 6:00 AM	Begins to fall

Peak melatonin levels occur during the middle of the night and are typically suppressed by early morning light.

4. Factors Affecting Melatonin Production

- **Light exposure:** Blue light from screens (phones, laptops) can block melatonin secretion.
- **Age:** Melatonin levels naturally decrease with age.
- **Shift work or jet lag:** Disrupts normal melatonin rhythms, leading to sleep disorders.
- **Medications or caffeine:** Certain substances can suppress melatonin release.

5. Disorders Related to Melatonin Disruption

When melatonin levels are out of sync, people may suffer from:

- **Insomnia** (trouble falling or staying asleep)
- **Delayed Sleep Phase Syndrome (DSPS)** – common in teens who sleep late and wake up late
- **Jet Lag** – temporary disorder caused by traveling across time zones
- **Shift Work Sleep Disorder** – affects people who work night shifts
- **Non-24-Hour Sleep-Wake Disorder** – mostly in blind individuals, where melatonin secretion is not properly synchronized

MELATONIN IN ADOLESCENTS AND SLEEP DELAY:

1. Introduction: Sleep in Adolescents

Adolescence is a period of significant physical, emotional, and neurological development. One of the most noticeable biological changes is a shift in sleep patterns. Many teenagers tend to stay awake late at night and struggle to wake up early in the morning. This behavior is not just a matter of choice—it is **biologically driven**, in part by changes in melatonin secretion.

2. Delayed Sleep Phase Syndrome (DSPS) in Teenagers

During adolescence, the timing of melatonin secretion naturally **shifts later into the evening**, a phenomenon known as **Delayed Sleep Phase Syndrome (DSPS)** or **delayed circadian rhythm**.

- In children, melatonin production starts around **7–8 PM**.
- In adolescents, it may not begin until **10–11 PM or later**, leading to a delay in feeling sleepy.
- Despite this delay, school start times often remain early (around 7–8 AM), causing a **chronic sleep deficit**.

3. Consequences of Sleep Delay in Teens

This mismatch between biological sleep cycles and social demands has several effects:

- **Daytime sleepiness**
- **Poor academic performance**
- **Irritability and mood swings**
- **Increased risk of depression and anxiety**
- **Reduced attention span and memory problems**

Sleep deprivation in teens can have both immediate effects and long-term health implications.

4. Role of Melatonin in Managing Sleep Delay

Melatonin supplements, when properly timed, can be used to **reset the sleep cycle** in teenagers with delayed sleep phase. This involves:

- **Taking low-dose melatonin (0.5–3 mg) 1–2 hours before desired bedtime.**
- **Combining with good sleep hygiene:** limiting screen time, dimming lights in the evening, and keeping a consistent schedule.

Melatonin should only be used under **medical supervision**, especially in adolescents, to avoid side effects and improper timing that may worsen sleep problems.

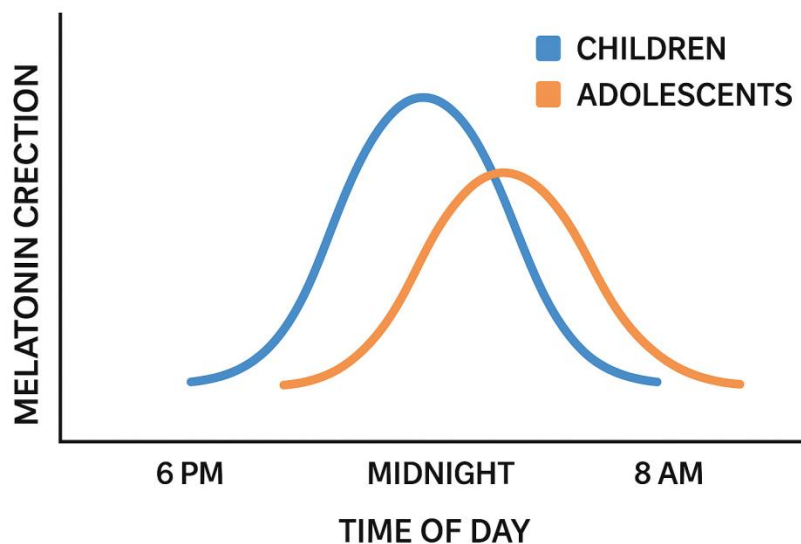
5. Lifestyle Factors That Worsen the Delay

Several habits can amplify the natural delay in melatonin:

- **Excessive screen time** at night (phones, tablets, gaming).
- **Irregular sleep schedules**, especially sleeping late on weekends.
- **Caffeine consumption** in the evening.
- Lack of **daylight exposure** in the morning, which is important to reset the circadian clock.

6. Strategies to Improve Sleep in Teens (Without Medication)

- **Light therapy** in the morning to help shift the circadian rhythm earlier.
- **Reducing blue light exposure** in the evening (use of filters or screen curfew).
- Creating a **dark, quiet, and cool sleep environment**.
- Encouraging **consistent wake-up times**, even on weekends.



Melatonin and Sleep in Adults (Ages 25–57)

1. Melatonin Stability in Adulthood

Between the ages of **25 to around 55–60**, melatonin production is relatively **stable**. During this period:

- The **circadian rhythm** (biological clock) is usually well-regulated.
- Melatonin continues to be secreted in a **predictable pattern**—rising in the evening, peaking at night, and falling by morning.
- Adults generally experience **7–9 hours of consolidated sleep** per night when melatonin production is healthy.

2. Common Challenges Affecting Melatonin and Sleep in Adults

While the internal melatonin rhythm is intact, many adults experience **sleep disturbances** due to behavioral and environmental factors:

a) Chronic Stress and Anxiety

- Elevated stress hormones like cortisol can interfere with melatonin production.
- Adults juggling work, family, and social obligations often find it hard to unwind at night.

b) Irregular Sleep Schedules

- Shift work, long hours, or inconsistent routines can **desynchronize** the circadian rhythm.
- This affects the timing of melatonin release, leading to difficulty falling or staying asleep.

c) Screen Exposure and Blue Light

- Prolonged use of phones, tablets, and laptops in the evening **suppresses melatonin** by blocking signals that indicate darkness to the brain.
- Adults working late or watching TV before bed often delay melatonin onset, causing **delayed sleep**.

d) Diet, Caffeine, and Alcohol

- Stimulants like caffeine delay melatonin production if consumed late in the day.
- Alcohol, while initially sedating, **disrupts natural melatonin cycles** and causes fragmented sleep.

3. Sleep Disorders in Adults

Some adults between 25–57 experience clinically diagnosed sleep conditions, often linked to melatonin imbalance or poor circadian alignment:

- **Insomnia**
- **Delayed Sleep Phase Disorder**
- **Shift Work Sleep Disorder**
- **Obstructive Sleep Apnea** (though not directly related to melatonin, it co-occurs with poor sleep quality)

4. Supporting Healthy Melatonin Production

Adults can optimize melatonin and sleep by:

- **Maintaining a regular sleep-wake schedule.**
- **Avoiding screens** at least an hour before bed.
- Using **warm, dim lighting** in the evening.
- **Exposing themselves to natural light in the morning** to reinforce circadian rhythms.
- Avoiding caffeine after 2 PM and limiting alcohol close to bedtime.

In some cases, **low-dose melatonin supplements** (0.5–3 mg) may be used temporarily for jet lag, sleep phase issues, or shift work—but only under professional guidance.

In adults aged 25–57, melatonin rhythms are biologically stable, but modern lifestyles can disturb their effectiveness. Factors like stress, screen exposure, irregular schedules, and poor sleep hygiene can delay or suppress melatonin production, leading to sleep disturbances. Maintaining circadian-friendly habits can help ensure natural melatonin function and better sleep quality.

Effects of Artificial Light and Screen Time on Melatonin:

1. Light and the Human Body Clock

Our body's internal clock, or **circadian rhythm**, is regulated largely by **light exposure**. The brain interprets light signals through the **eyes** and adjusts the secretion of **melatonin**, the hormone responsible for promoting sleep. While natural sunlight helps us stay awake and alert during the day, **darkness triggers melatonin production** at night, preparing the body for rest.

However, in modern lifestyles, **artificial lighting**—especially from **electronic screens**—can disrupt this delicate balance.

2. How Light Affects Melatonin

- Melatonin secretion is **inhibited by light**, especially **blue light**, which has a wavelength of around 460–480 nm.
- The **retina** in the eye contains specialized cells (intrinsically photosensitive retinal ganglion cells, or ipRGCs) that detect blue light and signal the **suprachiasmatic nucleus (SCN)** in the brain to suppress melatonin.
- Even moderate light exposure at night can **delay the onset** of melatonin release, reduce its **overall levels**, and **shift the circadian clock**.

3. Role of Screens and Digital Devices

Screens from phones, tablets, laptops, and televisions emit a significant amount of **blue light**.

Negative impacts of screen time at night:

- **Delayed sleep onset:** Using screens before bed delays melatonin release, making it harder to fall asleep.
- **Reduced melatonin levels:** Prolonged exposure lowers the natural peak in melatonin levels.
- **Poor sleep quality:** Even after falling asleep, the quality and depth of sleep may be diminished.
- **Shorter sleep duration:** People using devices at night often sleep later and for fewer hours.

4. Scientific Evidence

- A Harvard study found that **blue light exposure suppresses melatonin twice as much** as other wavelengths and **shifts circadian rhythms by up to 3 hours**.
- A study in adolescents showed that **1 hour of smartphone use** in the evening significantly reduced melatonin onset compared to those not exposed to screens.

5. Behavioral Effects

Reduced melatonin due to screen time is linked to:

- **Fatigue and irritability** during the day
- **Decreased concentration and memory**
- **Mood disorders** (e.g., anxiety and depression)
- **Weakened immune function** (due to poor sleep)

6. Solutions and Mitigation

Here are steps to reduce the impact of artificial light and screens on melatonin:

Strategy	How It Helps
Use " night mode " or blue light filters	Reduces blue light emitted from screens
Install blue-light-blocking apps (e.g., f.lux)	Automatically adjusts screen color with time
Wear blue-blocking glasses	Filters out blue wavelengths from any source
Avoid screens 1–2 hours before bed	Gives melatonin time to rise naturally
Use dim, warm lighting in the evening	Mimics natural evening light
Expose yourself to bright light in the morning	Resets the circadian clock and promotes alertness

Natural vs. Synthetic Melatonin:

1. What Is Natural Melatonin?

a) Endogenous Melatonin

- Produced by the **pineal gland** in the brain, primarily at night.
- Synthesis is triggered by **darkness** and inhibited by **light**.
- Regulates the **circadian rhythm**, promotes **sleep onset**, and supports **antioxidant activity**.

b) Melatonin in Foods

Some foods contain small amounts of natural melatonin or its precursors (tryptophan and serotonin):

- **Tart cherries, grapes, tomatoes**
- **Nuts** (especially pistachios)
- **Milk**
- **Bananas, oranges**

These dietary sources may **support natural melatonin production**, but the levels are usually too low to cause sedative effects on their own.

2. What Is Synthetic Melatonin?

a) Manufactured in Laboratories

- Most supplements today contain **synthetic melatonin** made from **chemical synthesis**.
- Designed to **mimic the structure and function** of naturally produced melatonin.
- Available in **tablets, capsules, gummies, liquids**, and **extended-release** forms.

b) Controlled Dosage

- Doses usually range from **0.3 mg to 10 mg**, with 1–3 mg being typical for sleep support.
- Extended-release forms can help people who have trouble staying asleep, not just falling asleep.

3. Key Differences: Natural vs. Synthetic Melatonin

Feature	Natural Melatonin (Body/Food)	Synthetic Melatonin (Supplements)
Source	Pineal gland, foods	Chemically synthesized in labs
Regulation	Based on light/dark cycle	Taken externally, dosage controlled
Bioavailability	Low to moderate (from food)	High and consistent per dose
Safety	Endogenously regulated	Generally safe short-term, but may interfere with hormonal balance if overused
Use Cases	Maintains natural sleep rhythm	Used for insomnia, jet lag, shift work sleep disorder
Side Effects	None	May cause drowsiness, headaches, dizziness, or vivid dreams in some people

4. Are There Natural-Sourced Melatonin Supplements?

Some supplements claim to be derived from **natural sources**, such as:

- **Animal pineal glands** (e.g., bovine) – Rare today due to **safety and disease concerns**.
- **Plant-based melatonin** – Some companies extract melatonin from **plants like rice or herbs**, though the amounts are much lower and more expensive to produce.

Most “natural” melatonin on the market is still **synthetically manufactured**, but designed to **match the molecule** of natural melatonin exactly.

5. When to Use Synthetic Melatonin

Synthetic melatonin is often used in adults (including those aged 25–57) for:

- **Jet lag recovery**
- **Delayed Sleep Phase Syndrome**
- **Insomnia**
- **Shift work sleep disorder**
- Supporting sleep in conditions like **anxiety or stress-related insomnia**

It is generally **non-habit-forming**, but should be used **under guidance**, especially in people with hormonal or mood disorders.

Survey Analysis: Investigating Public Awareness and Habits Related to Melatonin and Sleep

As part of this investigatory biology project, I conducted a **survey using Google Forms** to explore the **knowledge, sleep habits, and screen exposure patterns** among individuals aged 25 to 57, with a focus on how these factors may influence **natural melatonin production** and **sleep quality**.

SLEEP AND SCREENS: A SURVEY BY AAYUSHI RAO

This survey was conducted by me. A Google Forms link was shared to family and friends and they were requested to fill out the form by answering the questions honestly and mindfully.

RESPONSES RECORDED: 73

DEMOGRAPHIC GROUP: I did not limit the responses to any specific age group. People aged between 13 to 57 have filled the form.

MODE: Google forms

SURVEY TOPICS: Section 1: Demographic data

Section 2: Sleep Habits

Section 3: Screen Time and Light Exposure

Section 4: Awareness and Behaviour

Section 5: Lifestyle

Here are screenshots of the questions:

The screenshot shows the Google Forms interface for a survey titled "SLEEP AND SCREENS: A BIOLOGY PROJECT SURVEY". The form is published and saved in Google Drive. The survey is divided into five sections, with the first section titled "Section 1 of 5" currently active. The survey introduction text reads: "This is for a class XII biology project on Melatonin and your honest answers will help real analysis. Please respond thoughtfully." The survey is attributed to "Aayushi Rao". The first question is a "Short answer text" question labeled "Name". The form has a teal background and a white question box.

Name

Short answer text

Age *

Short answer text

Gender *

☐ Male

☐ Female

Section 2 of 5

SLEEP HABITS



Description (optional)

What time do you usually go to bed on weekdays? *

☐ Before 9 PM

☐ 9–10 PM

☐ 10–11 PM

☐ After 11 PM

How many hours of sleep do you get on average each night? *

☐ Less than 5

☐ 5–6

☐ 6–7

☐ 7–8

☐ More than 8

Do you find it difficult to fall asleep at night? *

- ☐ Never
- ☐ Rarely
- ☐ Sometimes
- ☐ Often
- ☐ Always

On average, how long does it take you to fall asleep after going to bed? *

- ☐ Less than 15 minutes
- ☐ 15–30 minutes
- ☐ 30–60 minutes
- ☐ More than 1 hour

Do you feel refreshed when you wake up in the morning? *

- ☐ Yes
- ☐ No
- ☐ Sometimes

Section 3 of 5

SCREEN TIME AND LIGHT EXPOSURE:

Description (optional)

How many hours do you spend on screens (phone, computer, TV) per day? *

- ☐ Less than 2
- ☐ 2–4
- ☐ 4–6
- ☐ More than 6

Do you use screens **within 1 hour before bedtime?** *

☐ Yes

☐ No

What kind of screen activity do you usually engage in before bed? *

☐ Watching Videos

☐ Social Media

☐ Studying

☐ Gaming

☐ Other...

Do you use any kind of **night mode** on your devices? *

☐ Yes always

☐ Sometimes

☐ Never

Section 4 of 5

AWARENESS AND BEHAVIOUR



Description (optional)

Have you heard of melatonin before? *

☐ Yes

☐ No

Are you aware that screen light can affect sleep hormones like melatonin? *

- ☐ Yes
- ☐ No
- ☐ Maybe

Have you ever taken melatonin supplements or sleep aids? *

- ☐ Yes
- ☐ No

Section 5 of 5

LIFESTYLE



Description (optional)

Do you feel sleepy during daytime? *

- ☐ Never
- ☐ Sometimes
- ☐ Often
- ☐ Always

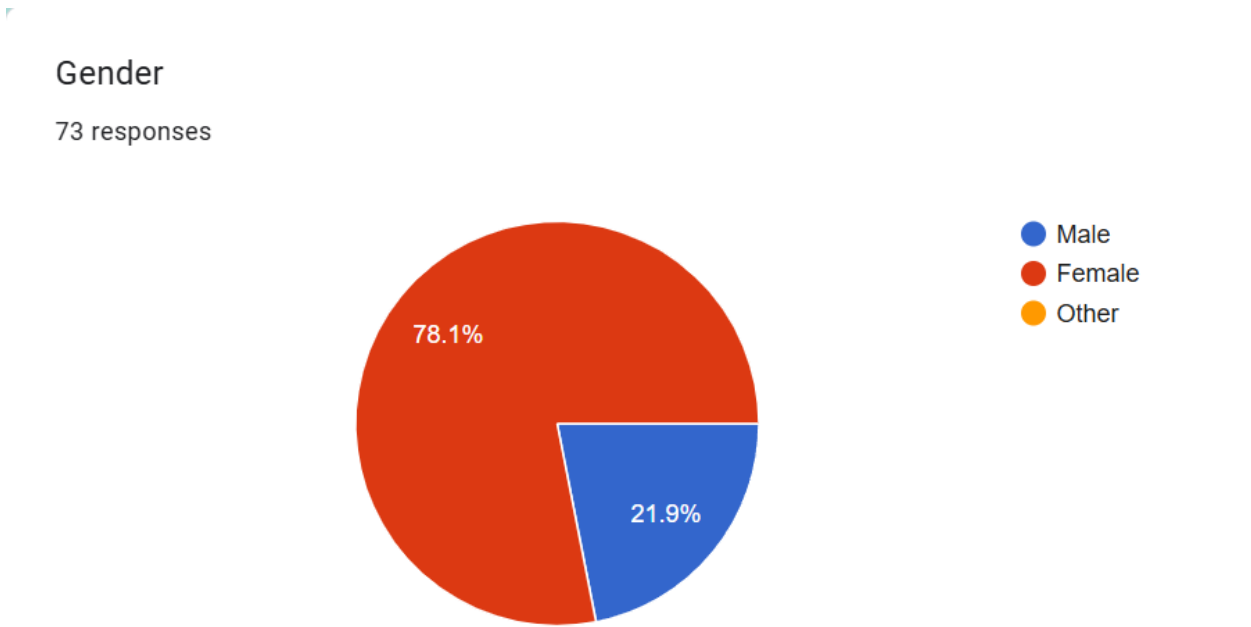
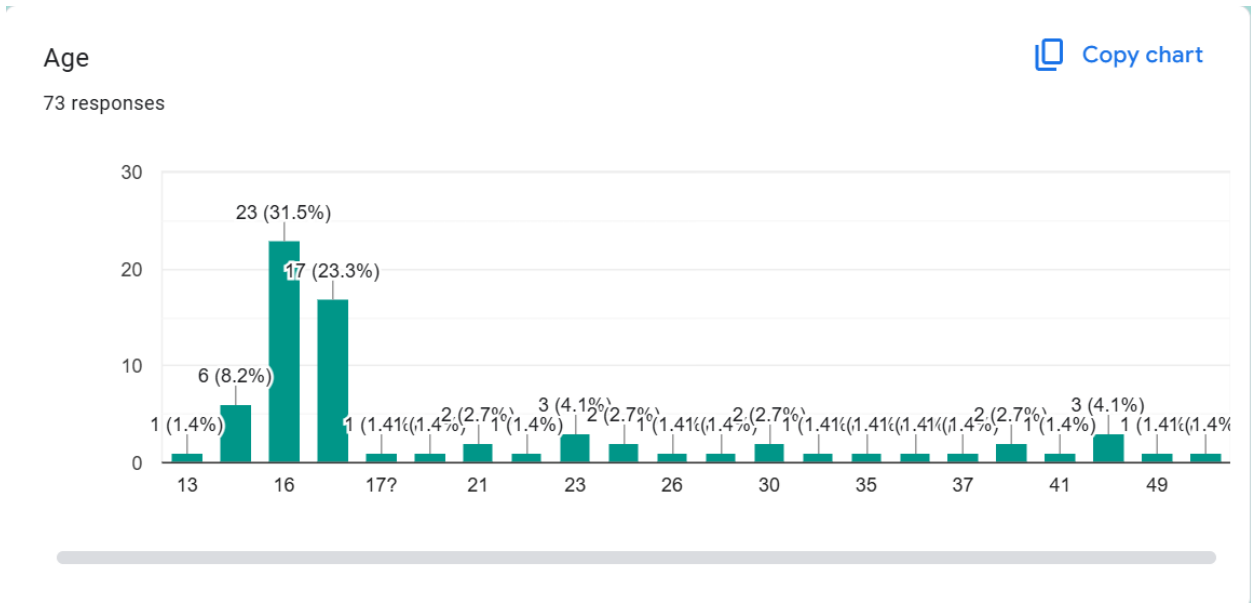
On a scale of 1–5, how would you rate your overall sleep quality in the past week? *

1 = Very Poor, 5 = Excellent

- | | | | | |
|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| 1 | 2 | 3 | 4 | 5 |
| ☐ | ☐ | ☐ | ☐ | ☐ |

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SURVEY ANALYSIS:



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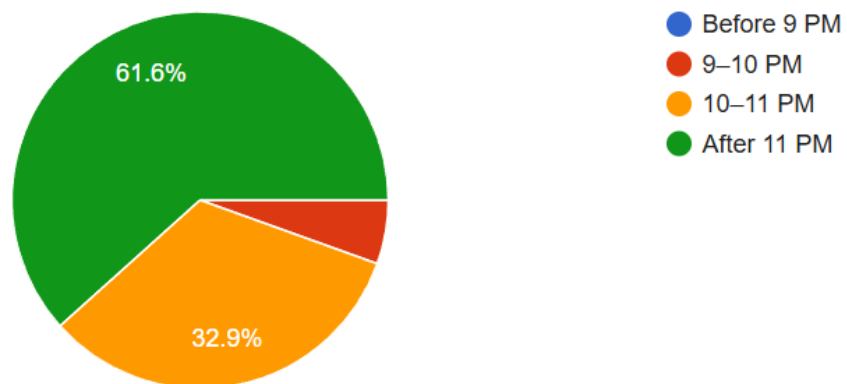
57 respondents: female

16 respondents: male

SLEEP HABITS

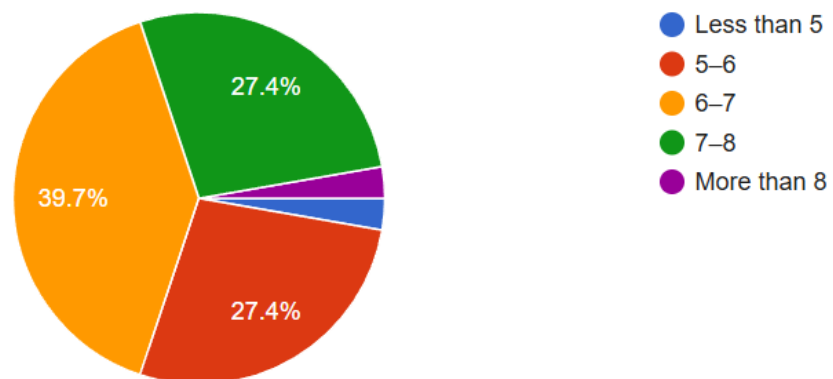
What time do you usually go to bed on weekdays?

73 responses



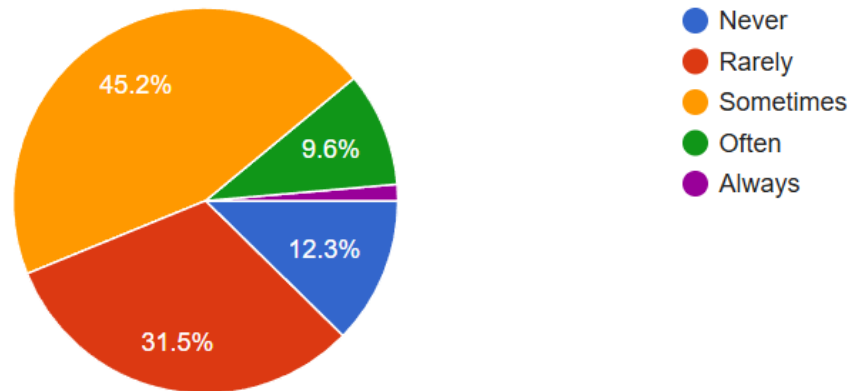
How many hours of sleep do you get on average each night?

73 responses



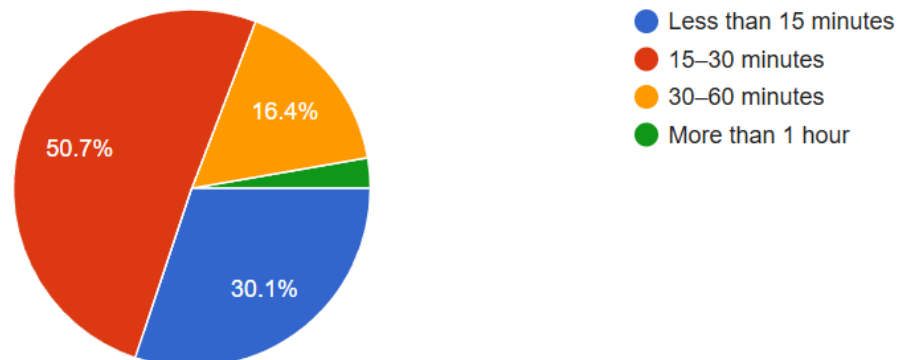
Do you find it difficult to fall asleep at night?

73 responses



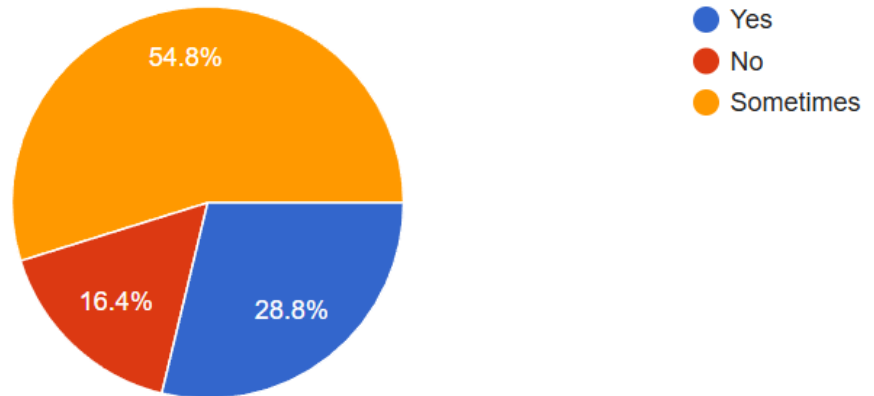
On average, how long does it take you to fall asleep after going to bed?

73 responses



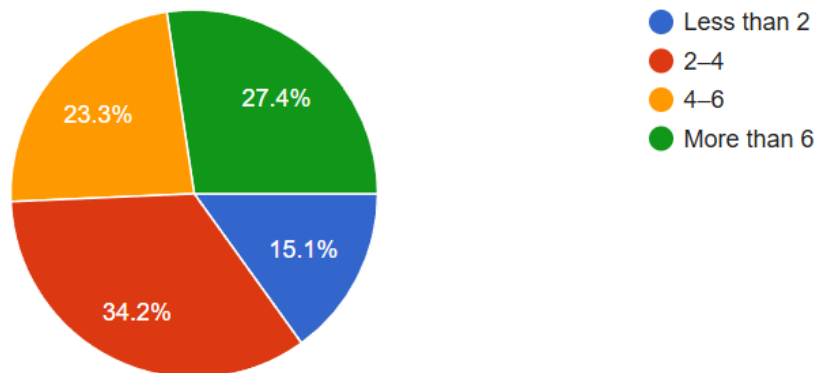
Do you feel refreshed when you wake up in the morning?

73 responses



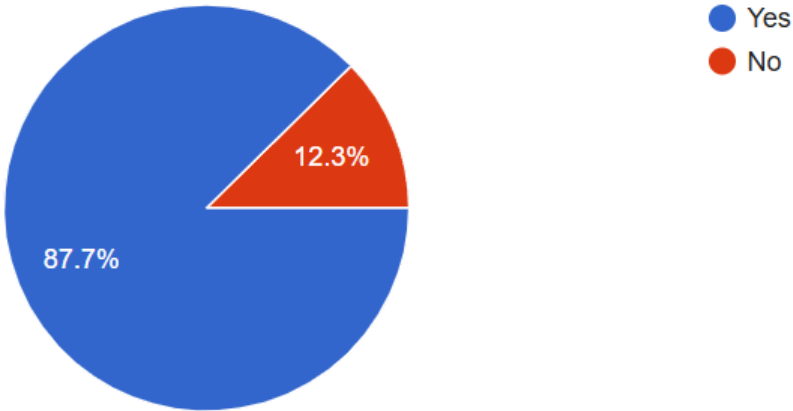
How many hours do you spend on screens (phone, computer, TV) **per day**?

73 responses



Do you use screens **within 1 hour before bedtime?**

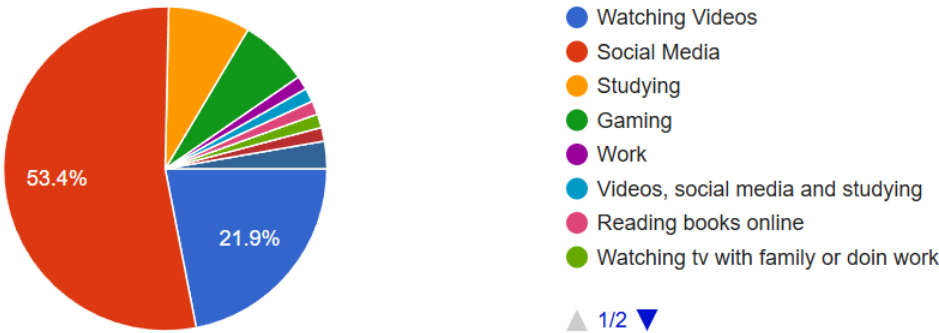
73 responses



What kind of screen activity do you usually engage in before bed?

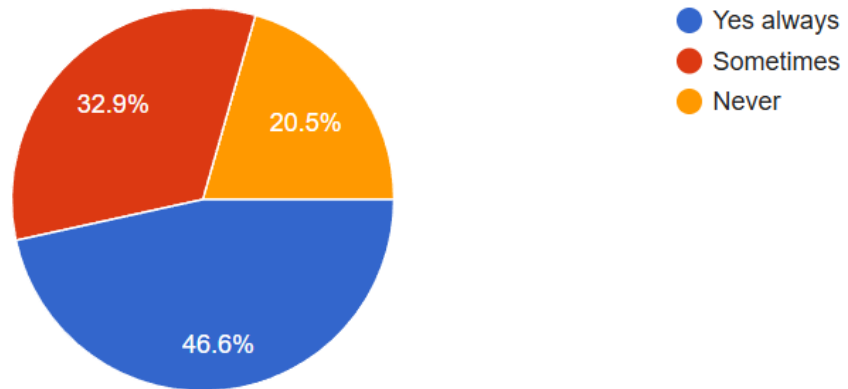
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73 responses



Do you use any kind of **night mode** on your devices?

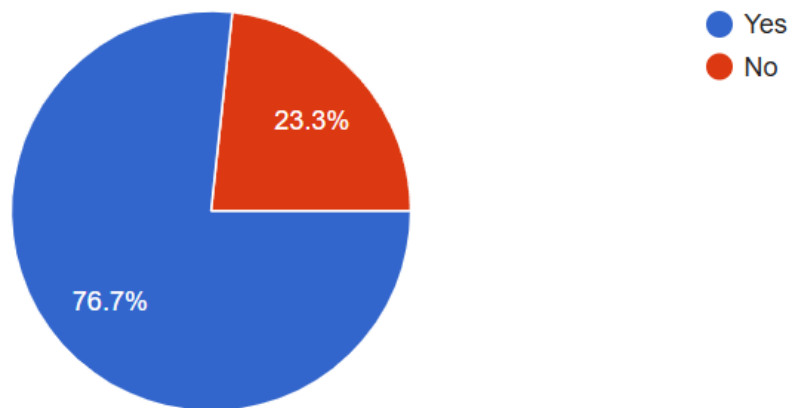
73 responses



AWARENESS AND BEHAVIOUR

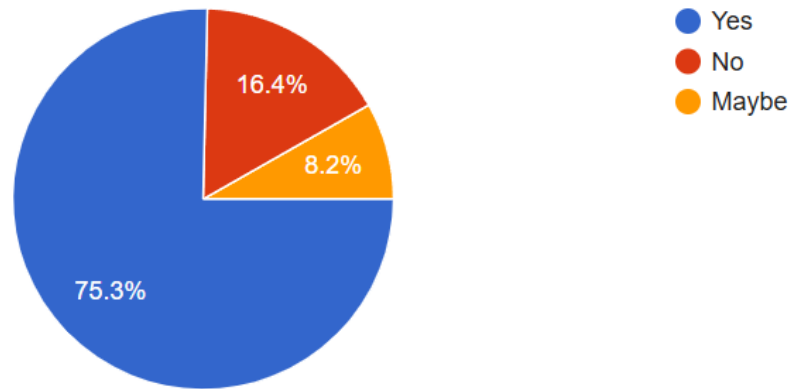
Have you heard of melatonin before?

73 responses



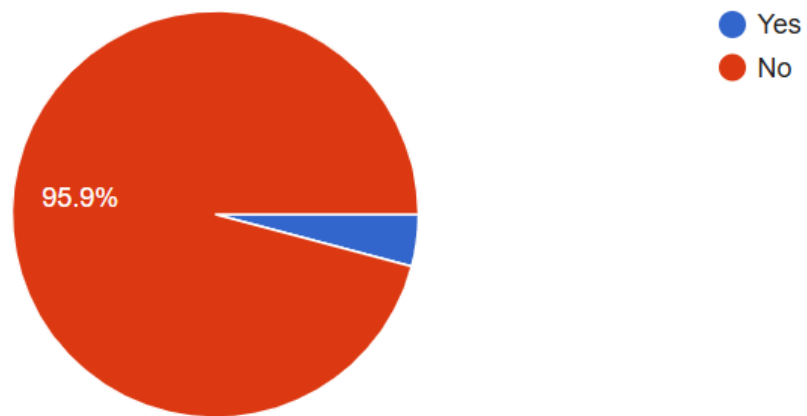
Are you aware that screen light can affect sleep hormones like melatonin?

73 responses



Have you ever taken melatonin supplements or sleep aids?

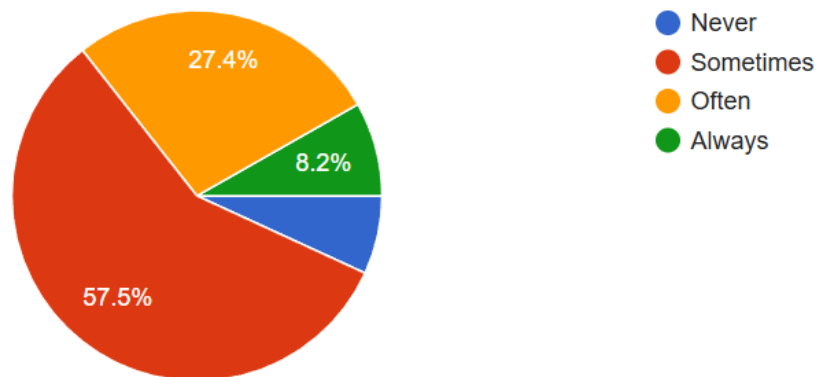
73 responses



LIFESTYLE

Do you feel sleepy during daytime?

73 responses

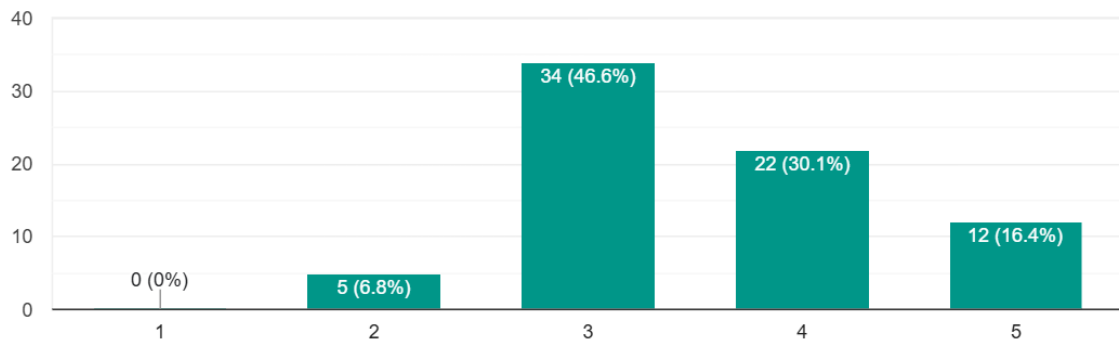


On a scale of 1–5, how would you rate your overall sleep quality in the past week?

[Copy chart](#)

1 = Very Poor, 5 = Excellent

73 responses



This analysis is based on general trends from the collected responses.

1. Sleep Patterns and Habits

- **Bedtime Trends:**

Around **32.9% of respondents** reported going to bed between **between 10 pm to 11pm**, which aligns reasonably well with the body's natural circadian rhythm.

However, a notable **61.6%** sleep after **11 PM**, indicating a delayed sleep schedule, possibly influenced by extended screen time or work patterns. Such delays can result in a mismatch between melatonin secretion and actual sleep onset, leading to sleep debt over time.

- **Sleep Duration:**

The majority i.e **67.1%** sleep between **6 to 8 hours** per night, within the recommended range for adults. However, **30.1%** sleep less than **6 hours**, which can lead to chronic fatigue, impaired concentration, and weakened immune function—symptoms often linked to **melatonin dysregulation**.

- **Sleep Onset & Difficulty:**

Approximately **56.2 %** reported **difficulty falling asleep**, and **19.1%** noted taking **more than 30 minutes** to fall asleep. This delay in sleep initiation is a strong indicator of **circadian disruption**, possibly linked to light exposure, stress, or irregular schedules.

- **Sleep Quality:**

On a 1–5 rating scale, most participants i.e **46.6%** chose **3 (average)** or lower, suggesting that even with adequate sleep duration, the quality of rest is lacking. Poor sleep quality often correlates with inconsistent melatonin rhythms or late-night screen habits.

2. Screen Exposure and Blue Light Impact

- **Total Daily Screen Time:**

A significant **50.7%** of respondents spend **over 4 hours** daily on screens (phones, computers, TVs). With digital work, entertainment, and social interaction now screen-centric, this level of exposure is almost unavoidable—but it comes at a cost to **melatonin balance**, especially if unfiltered blue light is involved.

- **Screens Before Bed:**

Nearly **87.7%** admitted to using screens within **1 hour before bedtime**—a critical period when melatonin should rise. Blue light from screens is known to **suppress melatonin production**, delay the onset of drowsiness, and interfere with the body's readiness for sleep. This finding is one of the clearest behavioral links to the sleep difficulties observed.

- **Night Mode Usage:**

46.6% use **night mode always** and 32.9% use night mode **sometimes**. Only a moderate proportion of respondents use **night mode or blue light filters**. While night mode can reduce blue light exposure, its inconsistent use limits its effectiveness in preserving melatonin rhythms.

3. Awareness and Supplement Use

- **Awareness of Melatonin:**

About **76.7%** of respondents reported having **heard of melatonin**, showing basic public familiarity. However, awareness did not always translate into understanding its function or how behavior affects its production.

- **Knowledge of Light's Role:**

Slightly more than **75.3%** were aware that **screen light affects melatonin**, indicating understanding of the biological consequences of screen habits.

4. Daytime Effects

- **Feeling Refreshed:**

Only **28.8%** of respondents consistently felt **refreshed in the morning**, while the rest reported grogginess or the need for stimulants like caffeine. This suggests that even when sleep occurs, it may not be **deep or restorative**, likely due to poor melatonin alignment with sleep phases.

- **Daytime Sleepiness:**

A total of 93.2% experience **daytime sleepiness**, a classic symptom of circadian misalignment and poor sleep quality. This can result in reduced productivity, mood fluctuations, and long-term health consequences.

CONCLUSION:

The survey highlights a clear connection between modern digital habits and disruption of natural sleep cycles, with melatonin playing a central role. While general awareness of melatonin exists, behavioral patterns such as late-night screen exposure and inadequate sleep hygiene continue to hinder its effectiveness. For healthier sleep outcomes, there is a growing need to prioritize lifestyle changes over reliance on supplements. By adopting practical measures—like reducing screen use before bed, maintaining consistent sleep schedules, and creating darker, tech-free sleep environments—individuals can support their body's natural melatonin production and enhance overall well-being. This project underscores the importance of aligning our habits with our biology to achieve truly restorative sleep.